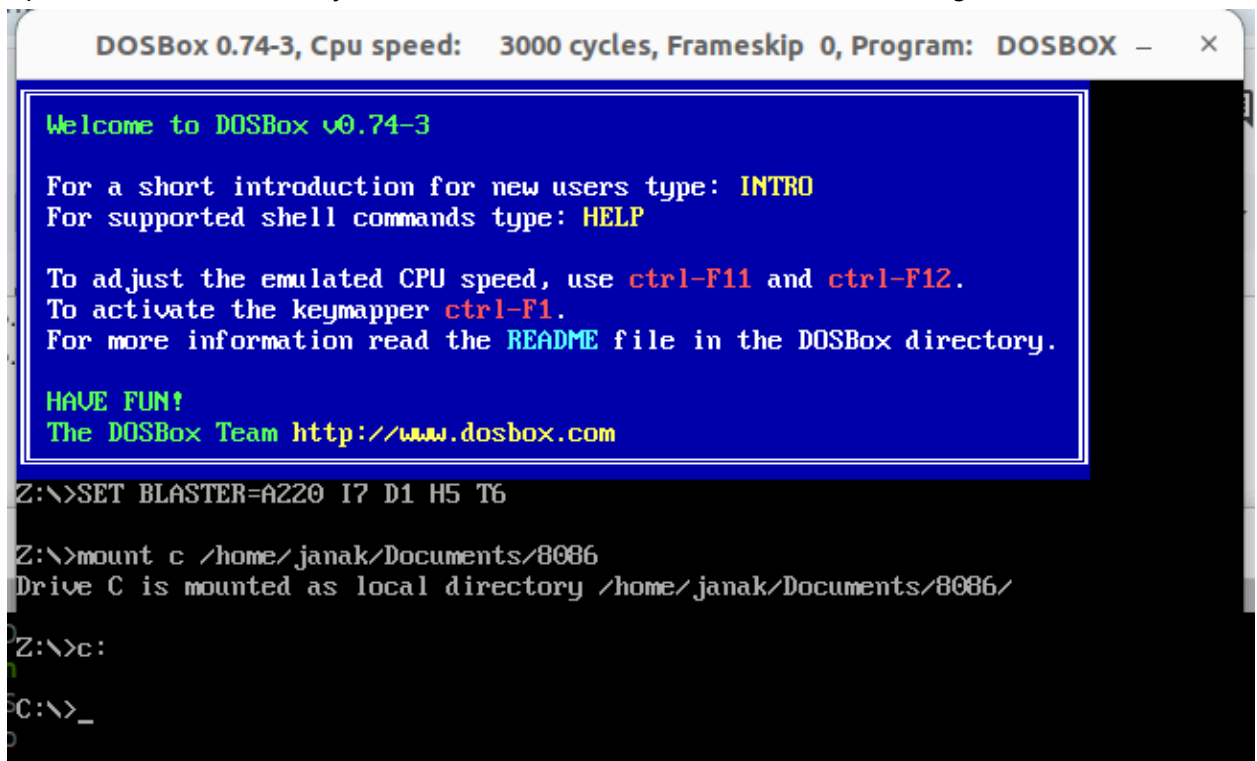


1. Open dosbox and mount your folder make sure that folder contains 'debug.exe'.



DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX

```
Welcome to DOSBox v0.74-3

For a short introduction for new users type: INTRO
For supported shell commands type: HELP

To adjust the emulated CPU speed, use ctrl-F11 and ctrl-F12.
To activate the keymapper ctrl-F1.
For more information read the README file in the DOSBox directory.

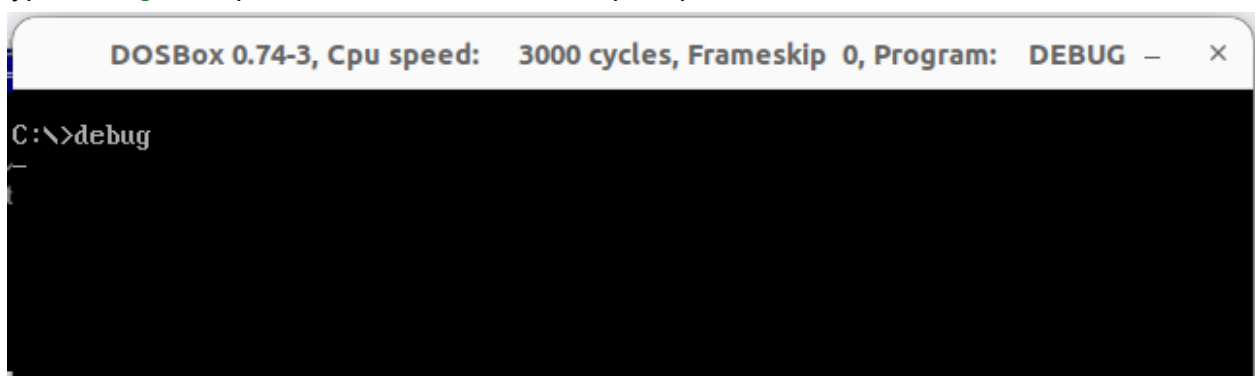
HAVE FUN!
The DOSBox Team http://www.dosbox.com

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount c /home/janak/Documents/8086
Drive C is mounted as local directory /home/janak/Documents/8086/

Z:\>c:
C:\>_
```

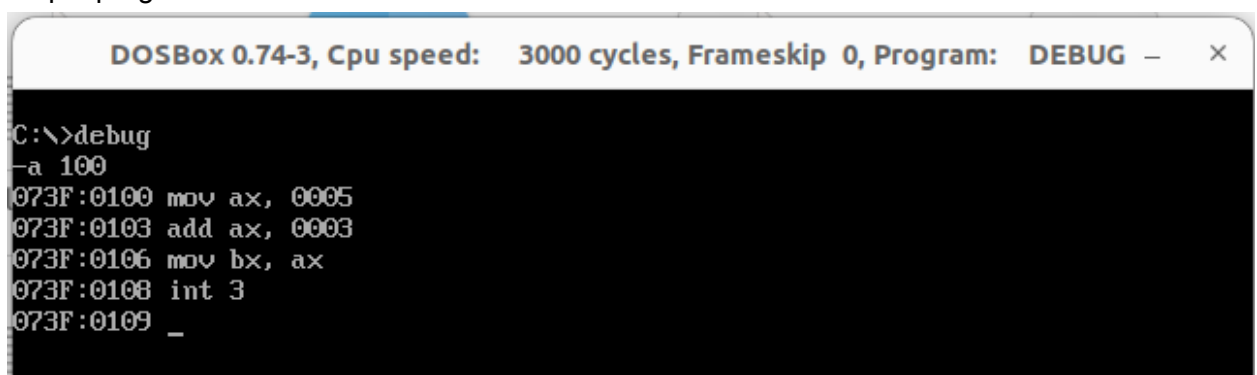
2. type **debug**, then press Enter. You will see a - prompt.



DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG

```
C:\>debug
-
```

3. Use the **a** (assemble) command starting at the default address (**CS:0100**) to write a simple program that adds two numbers:



DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG

```
C:\>debug
-a 100
073F:0100 mov ax, 0005
073F:0103 add ax, 0003
073F:0106 mov bx, ax
073F:0108 int 3
073F:0109 _
```

(Press **Enter** on a blank line to exit the assembler mode).

4. Use the **r** (register) command to view the initial register states:

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG - X
C:\>debug
-a 100
073F:0100 mov ax, 0005
073F:0103 add ax, 0003
073F:0106 mov bx, ax
073F:0108 int 3
073F:0109
-
-r
AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NV UP EI PL NZ NA PO NC
073F:0100 B80500      MOV     AX,0005
-
-
```

5. Use the **t** (trace) command step-by-step to execute each instruction and watch the values change in the registers (**AX**, **BX**, **IP**, etc.):

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG - X
C:\>debug
-a 100
073F:0100 mov ax, 0005
073F:0103 add ax, 0003
073F:0106 mov bx, ax
073F:0108 int 3
073F:0109
-
-r
AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NV UP EI PL NZ NA PO NC
073F:0100 B80500      MOV     AX,0005
-t
AX=0005 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0103  NV UP EI PL NZ NA PO NC
073F:0103 050300      ADD     AX,0003
-t
AX=0008 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0106  NV UP EI PL NZ NA PO NC
073F:0106 89C3      MOV     BX,AX
-
-
```

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG - x
073F:0103 add ax, 0003
073F:0106 mov bx, ax
073F:0108 int 3
073F:0109
-
-r
AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NV UP EI PL NZ NA PO NC
073F:0100 B80500          MOV     AX,0005
-t
AX=0005 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0103  NV UP EI PL NZ NA PO NC
073F:0103 050300          ADD     AX,0003
-t
AX=0008 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0106  NV UP EI PL NZ NA PO NC
073F:0106 89C3          MOV     BX,AX
-t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0108  NV UP EI PL NZ NA PO NC
073F:0108 CC          INT     3
```

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG - x
AX=0008 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0106  NV UP EI PL NZ NA PO NC
073F:0106 89C3          MOV     BX,AX
-t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0108  NV UP EI PL NZ NA PO NC
073F:0108 CC          INT     3
-t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00F7 BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=01A3 IP=13B1  NV UP DI PL NZ NA PO NC
01A3:13B1 55          PUSH    BP
-t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00F5 BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=01A3 IP=13B2  NV UP DI PL NZ NA PO NC
01A3:13B2 8BEC          MOV     BP,SP
-t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00F5 BP=00F5 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=01A3 IP=13B4  NV UP DI PL NZ NA PO NC
01A3:13B4 FF8E0200      DEC     WORD PTR [BP+0002]          SS:00F7=0109
```

In 8086 assembly language and the DOS DEBUG utility, **INT 3** (Interrupt 3) is a special one-byte software interrupt used primarily as a **breakpoint**.

- a. **What it does:** When the CPU executes an **INT 3** instruction, it pauses program execution and hands control back to the debugger or operating system.
 - b. **Why it's used:** In your DEBUG session, placing **INT 3** at the end of your code acts as a safe stopping point. It prevents the processor from blindly executing whatever random data or garbage memory happens to come next in the code segment.
6. To quit DEBUG and return to the DOS prompt, type:

```
AX=0008 BX=0008 CX=0000 DX=0000 SP=00F5 BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=01A3 IP=13B2  NU UP DI PL NZ NA PO NC
01A3:13B2 8BEC          MOV     BP,SP
-t
t
AX=0008 BX=0008 CX=0000 DX=0000 SP=00F5 BP=00F5 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=01A3 IP=13B4  NU UP DI PL NZ NA PO NC
01A3:13B4 FF8E0200      DEC     WORD PTR [BP+0002]      SS:00F7=0109
-q
C:\>_
```